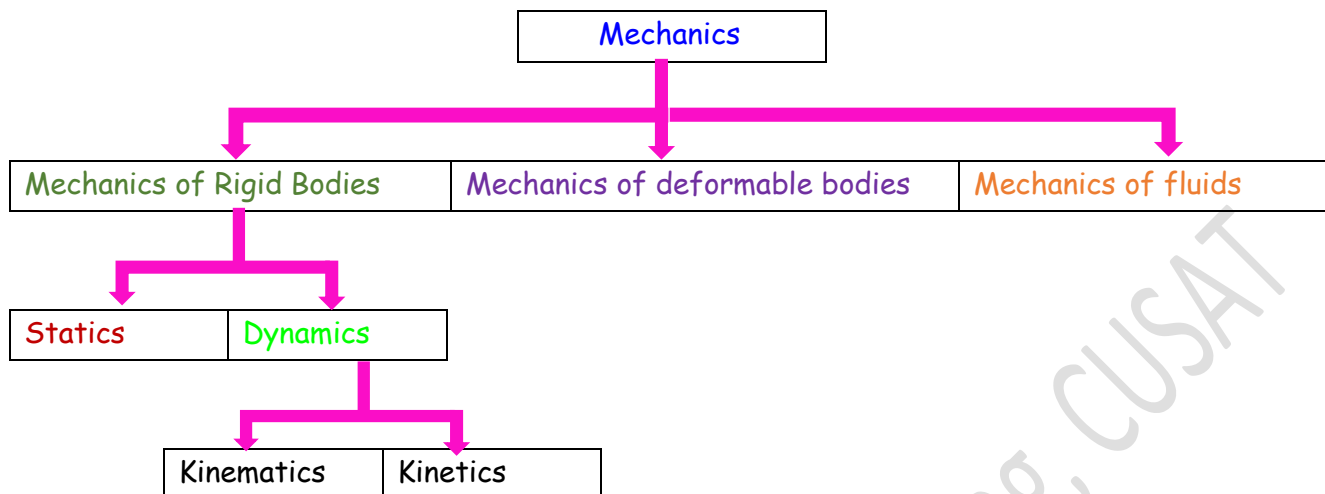


ENGINEERING MECHANICS

Mechanics deals with bodies or systems that are subjected to the action of forces. In general, Mechanics can be divided into three branches as below:



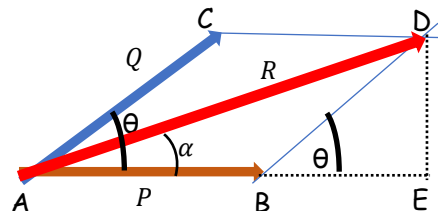
Rigid Body

All bodies will undergo some amount of deformation when forces are applied, however small it may be. In many situations our interest will be only on the state of the system (equilibrium or motion) rather than the deformation. In such cases, we can neglect the deformations occurring to the body due to the application of the forces. Then, the bodies are assumed to be rigid.

Principles of statics

1. Parallelogram law

If two forces P and Q are acting at point A at an angle θ are represented in magnitude and direction by the force vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, their resultant R represented by the force vector $\overrightarrow{AD}$ in magnitude and direction is given by the diagonal of the parallelogram $ABDC$.



From the $\triangle AED$,

$$AD^2 = AE^2 + ED^2 = (AB + BE)^2 + ED^2$$

From the right angled $\triangle BED$,

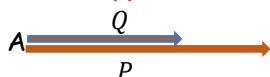
$$\sin \theta = \frac{ED}{BD} \ggg ED = BD \sin \theta = Q \sin \theta \quad \blacksquare \quad \cos \theta = \frac{BE}{BD} \ggg BE = BD \cos \theta = Q \cos \theta$$

$$R^2 = (P + Q \cos \theta)^2 + (Q \sin \theta)^2 = P^2 + 2PQ \cos \theta + Q^2 \cos^2 \theta + Q^2 \sin^2 \theta = P^2 + 2PQ \cos \theta + Q^2$$

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

$$\tan \alpha = \frac{ED}{AE} = \frac{Q \sin \theta}{P + Q \cos \theta}$$

case (i) $\theta = 0$



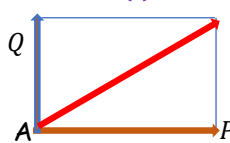
$$R = \sqrt{P^2 + Q^2 + 2PQ \cos 0}$$

$$R = P + Q$$

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta} = 0 \ggg$$

$$\alpha = 0$$

case (i) $\theta = 90$

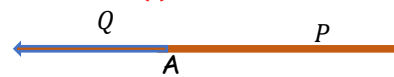


$$R = \sqrt{P^2 + Q^2 + 2PQ \cos 90}$$

$$R = \sqrt{P^2 + Q^2}$$

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta} = \frac{Q}{P} \gg \alpha = \tan^{-1} \frac{Q}{P}$$

case (i) $\theta = 180$



$$R = \sqrt{P^2 + Q^2 + 2PQ \cos 180}$$

$$R = P - Q$$

2. Equilibrium Law

Two forces can be in equilibrium (Resultant force = 0) only if they are equal, opposite, and collinear.

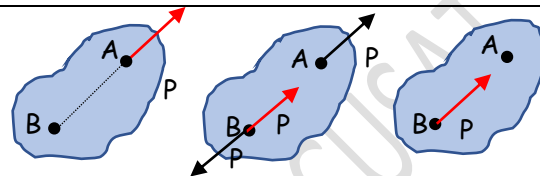
Look at the three cases of parallelogram law discussed above. We can see that resultant R can become zero, only for the third case when $P = Q$. For all other occasions, there will always be a resultant.

3. Principle of superposition

We can add to or subtract from a system of forces, **another force system in equilibrium** without any change in the state of the system.

4. Principle of transmissibility

A force can be transmitted **along its line of action** without any change in the **state** of the system



Consider a force P acting at point A on a body. Let B be a point on the body along the line of action of the force. Two equal and opposite forces of magnitude P are added at point B . This addition will not change the state of the system (principle of superposition). Now the two forces marked in black are equal, opposite, and collinear. Hence, they are in equilibrium which can be subtracted from the system (principle of superposition). Finally, we are left with only one force of magnitude P at B . That is, the force P at A is transmitted along its line of action.

5. Law of action and reaction

When a force is exerted on a rigid support, the support will exert an equal reaction in the opposite direction. A support is preventing the body from moving or rotating. **A support will exert a reaction force in a direction opposite to the direction of the motion prevented and a reaction moment in a direction opposite to the direction of the rotation prevented.**

Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

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